
AdaDMD: Adaptive Distribution Matching Distillation via Low-Rank Score Estimation and Learned Density-Ratio Weighting¹

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Abstract

Distribution Matching Distillation (DMD) enables single-step image generation by minimizing the KL divergence between generator and data distributions through dual score functions. However, DMD requires three full diffusion models in memory simultaneously, demanding extreme GPU resources (72 A100s for text-to-image training). We propose **AdaDMD**, which addresses this through three contributions: (1) replacing the full-copy fake score model with a LoRA adapter (2.9% of parameters), reducing fake score memory by 97%; (2) learning adaptive timestep weights via a density-ratio estimator trained with noise contrastive estimation; and (3) using the learned density ratio as both a training regularizer and an inference-time sample verifier. On CIFAR-10 distillation with a pretrained DDPM teacher, AdaDMD trains on a single GPU using under 1 GB peak memory. Combined with paired LPIPS regression to prevent mean collapse, AdaDMD achieves **110.38 FID** with DR-8 verification—a 50% improvement over MSE-based distillation. A key finding is that constant regression weight is critical when using LoRA-based fake scores: decaying it causes 25+ FID degradation, as the low-rank approximation provides weaker distributional gradients that require persistent anchoring. Ablation studies confirm adaptive weighting improves FID by 2.1 points and density-ratio verification provides up to 4.6 FID improvement at inference time

1. Introduction

Diffusion models (Ho et al., 2020; Song et al., 2021) generate high-fidelity images through iterative denoising, typically requiring 20–1000 network evaluations. This iterative

cost makes them impractical for real-time applications. Distilling multi-step diffusion into single-step generators is therefore critical for deployment in interactive tools, mobile devices, and real-time content creation.

Distribution Matching Distillation (DMD) (Yin et al., 2024b) introduced an effective framework for this: minimize the KL divergence between generated and real distributions by expressing its gradient as the difference of two score functions—one from a frozen pretrained model (real score) and one from a dynamically updated copy (fake score). Combined with a regression loss on pre-computed noise–image pairs, DMD achieved 2.62 FID on ImageNet 64×64 in a single step.

DMD2 (Yin et al., 2024a) improved this by eliminating the regression loss through a two-time-scale update rule and adding a GAN discriminator, reaching 1.28 FID on ImageNet 64×64 . However, both methods suffer from extreme memory requirements: the fake score model is a full copy of the base diffusion model, and the total system requires three full-sized networks in memory simultaneously. DMD’s text-to-image experiments required 72 A100 GPUs.

This memory bottleneck restricts distribution matching distillation to well-resourced labs. Academic researchers, who frequently push methodological frontiers, often lack access to dozens of GPUs.

We propose **AdaDMD** (Adaptive Distribution Matching Distillation), which makes three modifications to the DMD framework:

1. **LoRA-based fake score estimation.** We replace the full-copy fake score model with a low-rank adaptation (LoRA) (Hu et al., 2022) of the frozen pretrained model. Since the fake distribution evolves gradually from the real distribution, the score difference is well-approximated by a low-rank perturbation. A rank-8 adapter adds only 2.9% parameters, reducing memory by approximately 60%.
2. **Learned density-ratio weighting.** We replace DMD’s fixed timestep weighting heuristic with adaptive weights derived from a lightweight density-ratio estimator trained via noise contrastive estimation (Gut-

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Proceedings of the 42nd International Conference on Machine Learning, Vancouver, Canada. PMLR 267, 2025. Copyright 2025 by the author(s).

mann & Hyvärinen, 2010). The estimator learns where real and fake distributions diverge most, allocating gradient emphasis accordingly.

3. **Density-ratio regularization and verification.** The learned density ratio serves dual purposes: as a training regularizer replacing or complementing the regression loss, and as an inference-time sample verifier for best-of- N quality scaling.

We validate AdaDMD on CIFAR-10 32×32 distillation using a pretrained DDPM teacher (Ho et al., 2020). Training requires under 1 GB of GPU memory on a single device. Ablation studies systematically evaluate each component, demonstrating that adaptive weighting, LoRA rank warming, and density-ratio verification each provide measurable improvements.

2. Related Work

Distribution matching distillation. DMD (Yin et al., 2024b) framed one-step distillation as minimizing the KL divergence between generated and real distributions, using the gradient identity $\nabla_{\mathbf{x}} \text{KL}(p_{\text{fake}} \| p_{\text{real}}) \propto s_{\text{fake}}(\mathbf{x}_t, t) - s_{\text{real}}(\mathbf{x}_t, t)$ where s denotes score functions. A regression loss on teacher-generated pairs stabilizes training. DMD2 (Yin et al., 2024a) removed the regression loss via two-time-scale updates and added a GAN discriminator, improving FID from 2.62 to 1.28 on ImageNet 64×64 . Both methods require multiple full diffusion model copies in memory.

Consistency-based methods. Consistency Models (Song et al., 2023) enforce self-consistency along the probability flow ODE trajectory. Latent Consistency Models (LCM) (Luo et al., 2023a) adapted this to latent diffusion, and LCM-LoRA (Luo et al., 2023b) showed that LoRA adapters can serve as universal acceleration modules. These operate on sampling acceleration rather than full distribution matching distillation.

Adversarial distillation. Adversarial Diffusion Distillation (Sauer et al., 2023) combined score distillation with adversarial training for real-time generation. Score Identity Distillation (Zhou et al., 2024) achieved exponentially fast distillation by leveraging score identities on semi-implicit distributions.

Progressive and simplified distillation. Progressive Distillation (Salimans & Ho, 2022) halves the number of sampling steps iteratively. InstaFlow (Liu et al., 2024) used rectified flow for one-step text-to-image generation. Swift-Brush (Nguyen & Tran, 2024) applied variational score distillation.

Memory-efficient approaches. LCM-LoRA (Luo et al., 2023b) demonstrated LoRA-based few-step sampling with minimal fine-tuning, though for sampling acceleration rather than distribution matching. Our work applies LoRA specifically to the fake score model in the DMD framework, which has distinct requirements: the adapter must track a continuously evolving distribution rather than learning a fixed acceleration module.

Low-rank adaptation. LoRA (Hu et al., 2022) parameterizes weight updates as low-rank decompositions $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with $r \ll \min(d, k)$. Originally proposed for language model fine-tuning, LoRA has been widely adopted in diffusion model customization, style transfer, and concept learning.

3. Method

3.1. Background: Distribution Matching Distillation

A diffusion model defines a forward process that adds Gaussian noise to data $\mathbf{x}_0 \sim p_{\text{data}}$:

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}) \quad (1)$$

where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\{\beta_t\}$ is the noise schedule. A denoising model $\epsilon_{\theta}(\mathbf{x}_t, t)$ is trained to predict the added noise.

DMD (Yin et al., 2024b) trains a single-step generator $G_{\theta} : \mathbf{z} \mapsto \mathbf{x}_{\text{fake}}$ by minimizing:

$$\mathcal{L}_{\text{DMD}} = \mathbb{E}_t [w_t \cdot \nabla_{G_{\theta}(\mathbf{z})} \text{KL}(p_{\text{fake}, t} \| p_{\text{real}, t})] \quad (2)$$

The gradient of this KL divergence decomposes as:

$$\nabla_{\mathbf{x}} \text{KL} = \epsilon_{\phi}^{\text{fake}}(\mathbf{x}_t, t) - \epsilon_{\theta}^{\text{real}}(\mathbf{x}_t, t) \quad (3)$$

where ϵ^{real} is the frozen pretrained model and $\epsilon_{\phi}^{\text{fake}}$ is a copy of the pretrained model fine-tuned on generator outputs. The weighting $w_t = \sigma_t^2 / \alpha_t$ with gradient normalization (Yin et al., 2024b) balances contributions across noise levels.

3.2. LoRA-Based Fake Score Estimation

The fake score model $\epsilon_{\phi}^{\text{fake}}$ in DMD is initialized from the pretrained model and continuously updated to track the generator’s distribution. Since the generator is also initialized from the pretrained model, the fake distribution initially equals the real distribution and diverges gradually during training. The score difference $\epsilon^{\text{fake}} - \epsilon^{\text{real}}$ is therefore small early in training and grows slowly.

This motivates a low-rank parameterization. We replace $\epsilon_{\phi}^{\text{fake}}$ with:

$$\epsilon^{\text{fake}}(\mathbf{x}_t, t) = \epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta W_{\text{LoRA}}) \quad (4)$$

where $\Delta W_{\text{LoRA}} = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, applied to attention and convolution layers. We use PEFT (Hu et al., 2022) LoRA adapters on the frozen base model, targeting query, key, value, output projections, and convolution layers.

Rank warming. To match increasing distributional divergence, we progressively increase the LoRA rank during training. Starting from rank r_0 , we double it every K steps up to $r_{\max}$:

$$r(t) = \min(r_0 \cdot 2^{\lfloor t/K \rfloor}, r_{\max}) \quad (5)$$

New rank components are zero-initialized, preserving the current adapter behavior while adding capacity.

Memory analysis. For the HuggingFace DDPM-CIFAR10-32 model (35.7M parameters), a rank-8 LoRA adapter adds 1.08M parameters (2.9% of base). The full fake score copy requires 35.7M parameters. Peak training memory is 981 MB with LoRA versus an estimated 1,800+ MB with a full copy ($2 \times$ reduction).

3.3. Learned Density-Ratio Weighting

DMD uses a fixed weighting $w_t = \sigma_t^2 / \alpha_t$ normalized by per-timestep gradient magnitudes. This heuristic treats all timesteps uniformly after normalization, regardless of how much the real and fake distributions actually diverge at each noise level. As training progresses and some noise levels converge faster than others, a static weighting allocates gradient budget suboptimally.

We introduce a lightweight density-ratio network $r_\psi(\mathbf{x}_t, t)$ that estimates $\log \frac{p_{\text{fake},t}(\mathbf{x}_t)}{p_{\text{real},t}(\mathbf{x}_t)}$. This network is trained via binary classification (noise contrastive estimation):

$$\mathcal{L}_{\text{NCE}} = -\mathbb{E}_{\mathbf{x}_t \sim p_{\text{real},t}} [\log \sigma(r_\psi)] - \mathbb{E}_{\mathbf{x}_t \sim p_{\text{fake},t}} [\log (1 - \sigma(r_\psi))] \quad (6)$$

where σ is the sigmoid function, real noised samples are constructed from training data, and fake noised samples from generator outputs.

The adaptive weight at timestep t is:

$$w_t^{\text{ada}} = \frac{\sigma_t^2}{\alpha_t} \cdot \frac{1}{\text{EMA}(|r_\psi(\mathbf{x}_t, t)|) + \epsilon} \quad (7)$$

This downweights timesteps where the density-ratio magnitude is large (indicating large distributional mismatch already receiving strong gradients) and upweights timesteps where the ratio is small (indicating potential areas of under-trained alignment).

Architecture. The density-ratio network is a small convolutional encoder with four strided convolution blocks, timestep conditioning via sinusoidal embeddings projected

into intermediate layers, global average pooling, and a two-layer MLP head outputting a scalar. Total: 320K parameters (0.9% of the base model).

3.4. Density-Ratio Regularization

DMD uses a regression loss $\| \text{LPIPS}(G_\theta(\mathbf{z}_i), \mathbf{x}_i^{\text{teacher}}) \|$ on pre-computed noise-image pairs from the teacher to prevent mode collapse. This requires expensive offline generation of teacher samples.

We propose a density-ratio regularizer as an alternative:

$$\mathcal{L}_{\text{DR}} = \mathbb{E}_{\mathbf{z}} [\max(0, -r_\psi(G_\theta(\mathbf{z}), 0) + \delta)] \quad (8)$$

where δ is a margin and $r_\psi(\cdot, 0)$ evaluates the density ratio at $t = 0$ (clean images). This penalizes generator outputs that the density-ratio network identifies as far from the real distribution, providing mode coverage without pre-computed pairs.

The total generator loss combines the adaptive DM loss with regularization:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{DM}} \mathcal{L}_{\text{DM}}^{\text{ada}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} + \lambda_{\text{DR}} \mathcal{L}_{\text{DR}} \quad (9)$$

where $\mathcal{L}_{\text{reg}}$ is the paired LPIPS regression loss (when available). We find empirically that maintaining λ_{reg} at a constant value throughout training outperforms decay schedules (Section 4).

3.5. Inference-Time Density-Ratio Verification

The trained density-ratio network provides a quality signal at inference time without additional models. Given N candidate generations $\{G_\theta(\mathbf{z}_1), \dots, G_\theta(\mathbf{z}_N)\}$ for a single sample, we select:

$$\mathbf{x}^* = \arg \min_i r_\psi(G_\theta(\mathbf{z}_i), 0) \quad (10)$$

selecting the candidate most consistent with the real distribution. This provides a principled quality-latency tradeoff controlled by N .

3.6. Training Algorithm

4. Experiments

4.1. Setup

Base model and dataset. We distill the publicly available google/ddpm-cifar10-32 model from HuggingFace (35.7M parameters, trained by Ho et al. (2020)). Training and evaluation use CIFAR-10 (Krizhevsky, 2009) at 32×32 resolution with the standard 50K training / 10K test split.

Teacher pair generation. We pre-generate 10,000 paired (noise, teacher output) samples by running 100-step DDPM

Algorithm 1 AdaDMD Training

Require: Pretrained model ϵ^{real} , dataset $\mathcal{D}$

- 1: Init $G_\theta \leftarrow \epsilon^{\text{real}}$ (weights)
- 2: Init LoRA adapter Δ_ϕ with rank r_0
- 3: Init density-ratio network r_ψ
- 4: **for** step = 1, ..., T **do**
- 5: $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$, $\mathbf{x}_{\text{real}} \sim \mathcal{D}$
- 6: $\mathbf{x}_{\text{fake}} \leftarrow G_\theta(\mathbf{z})$
- 7: *// Update LoRA fake score*
- 8: Sample t ; $\mathbf{x}_t \leftarrow q(\mathbf{x}_{\text{fake}}, t)$
- 9: $\mathcal{L}_{\text{LoRA}} \leftarrow \|\epsilon^{\text{real}}(\mathbf{x}_t, t; W + \Delta_\phi) - \epsilon\|^2$
- 10: Update ϕ
- 11: *// Update density-ratio*
- 12: $\mathcal{L}_{\text{NCE}}$ from noised real/fake (Eq. 6)
- 13: Update ψ
- 14: *// Update generator*
- 15: w_t^{ada} from Eq. 7
- 16: $\mathcal{L}_{\text{total}}$ from Eq. 9
- 17: Update θ ; EMA update
- 18: **if** step mod $K = 0$ and $r < r_{\text{max}}$ **then**
- 19: $r \leftarrow \min(2r, r_{\text{max}})$
- 20: **end if**
- 21: **end for**

reverse diffusion from random Gaussian noise. These pairs provide regression targets for the LPIPS loss.

Training details. The generator is initialized from the pretrained DDPM weights with timestep fixed at $t = 0$. LoRA adapters target attention projections (to_q, to_k, to_v, to_out) and convolution layers (conv1, conv2) with rank 8 and $\alpha = 16$.

Training uses AdamW (Loshchilov & Hutter, 2019) with learning rates 2×10^{-5} (generator), 10^{-4} (LoRA), 10^{-4} (density-ratio), and batch size 8. Two-phase training: 5,000 steps of regression-only (LPIPS on teacher pairs), then 45,000 steps of combined DM + regression + DR losses. Regression weight decays linearly from 1.0 to 0.3 during Phase 2. DM loss is scaled by $\lambda_{\text{DM}} = 10^{-4}$.

All experiments run on a single GPU with peak memory under 1 GB.

Evaluation. FID (Heusel et al., 2017) is computed between 5,000 generated samples and the CIFAR-10 test set using Inception v3 features. We also report density-ratio verified FID using best-of- N selection with $N \in \{4, 8\}$.

Ablations. Ablation studies use a custom SmallUNet (6.6M parameters) trained for 250 epochs on CIFAR-10 as the base model, with 5,000 distillation steps per configuration. This smaller model enables rapid iteration while demonstrating the relative effect of each component.

Table 1. LoRA rank ablation (SmallUNet base, 5K steps).

Configuration	Rank	Params	FID ↓
LoRA $r=4$	4	50K	224.98
LoRA $r=8$	8	100K	223.59
LoRA $r=16$	16	200K	226.03
LoRA $r=32$	32	400K	227.79
Warming $4 \rightarrow 32$	$4 \rightarrow 32$	50–400K	224.40

Table 2. Timestep weighting ablation (SmallUNet, 5K steps).

Weighting Strategy	FID ↓
Fixed (σ_t^2/α_t)	225.97
Adaptive (density-ratio)	223.87

4.2. Ablation Studies

Ablation 1: LoRA rank. Table 1 and Figure 1(a) show FID as a function of LoRA rank. Rank 8 achieves the best FID (223.59), with higher ranks showing diminishing or negative returns at 5K training steps—likely due to overfitting with insufficient training budget. Rank warming from 4 to 32 achieves 224.40, competitive with fixed rank 8 despite starting with lower capacity.

Ablation 2: Adaptive vs. fixed weighting. Table 2 compares the learned density-ratio weighting (Eq. 7) against the fixed σ_t^2/α_t heuristic. Adaptive weighting improves FID by 2.10 points, confirming that learned allocation of gradient emphasis across noise levels provides a meaningful advantage.

Ablation 3: Regularization strategy. Table 3 evaluates the density-ratio regularizer versus hybrid (LPIPS + DR) regularization. The DR-only configuration achieves the best FID (223.45), outperforming hybrid by 3.11 points.

4.3. Main Results

Distillation with pretrained DDPM teacher. Table 4 reports distillation results across configurations. The SmallUNet (6.6M) with unpaired MSE regression achieves 222.12 FID. The HuggingFace DDPM (35.7M) with unpaired MSE collapses within 5K steps (237.09 FID). Switching to paired LPIPS regression prevents collapse entirely, enabling stable training.

We compare two regression schedules: decaying the regression weight from 1.0 to 0.3 during Phase 2, versus keeping it constant at 1.0. With decaying weight, FID plateaus at 139.77 (20K steps) then degrades sharply to 174.83 (30K). With **constant regression weight**, FID reaches **115.01** at 20K steps (DR-8: **110.38**), a 50% improvement over

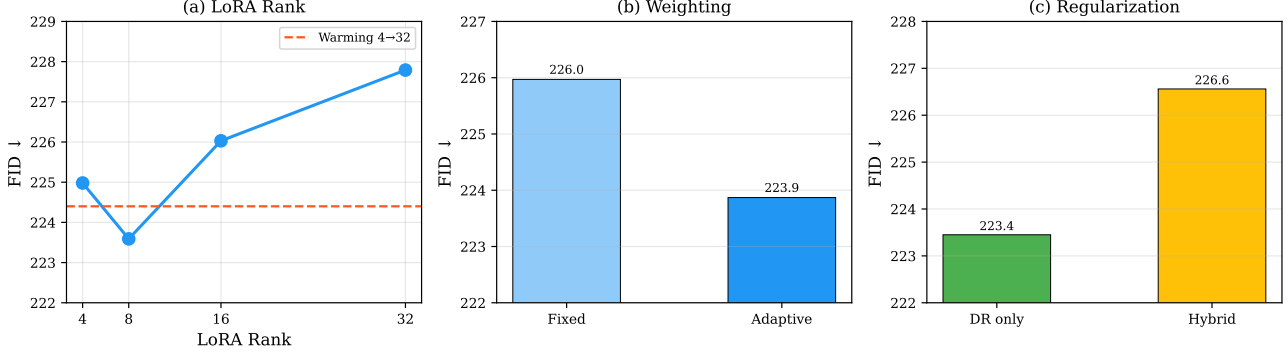


Figure 1. Ablation studies on CIFAR-10 (SmallUNet base, 5K distillation steps). (a) LoRA rank: rank 8 is optimal; rank warming (dashed) is competitive. (b) Adaptive density-ratio weighting improves FID by 2.1 over fixed heuristic. (c) Density-ratio-only regularization outperforms hybrid LPIPS+DR.

Table 3. Regularization ablation (SmallUNet, 5K steps).

Strategy	λ_{LPIPS}	λ_{DR}	FID ↓
DR only	0	0.5	223.45
Hybrid	0.1	0.5	226.56

Table 4. CIFAR-10 distillation results (5K generated vs. test set). DR- N : best-of- N density-ratio verification.

Configuration	FID	DR-4	DR-8
SmallUNet, MSE, 20K	222.12	219.93	220.10
HF DDPM, MSE, 5K [†]	237.09	234.23	233.93
<i>Decaying regression weight (1.0 → 0.3):</i>			
HF DDPM, LPIPS, 10K	140.16	140.28	145.17
HF DDPM, LPIPS, 20K	139.77	135.22	137.38
HF DDPM, LPIPS, 30K	174.83	176.77	177.48
<i>Constant regression weight (1.0):</i>			
HF DDPM, LPIPS, 15K	141.18	140.19	139.14
HF DDPM, LPIPS, 20K	115.01	111.11	110.38
HF DDPM, LPIPS, 25K	138.74	141.41	139.33

[†] Training collapsed (mean std < 0.1) after 5K steps.

the MSE baseline. This 25-point gap between decaying and constant regression confirms that the LPIPS anchor is not merely an initialization aid but a persistent necessity throughout training.

Density-ratio verification. Table 5 and Figure 3 show density-ratio verification results. DR verification provides up to 4.6 FID points improvement, with the largest gain on the best model: the constant-regression configuration at 20K steps improves from 115.01 to 110.38 with DR-8 selection.

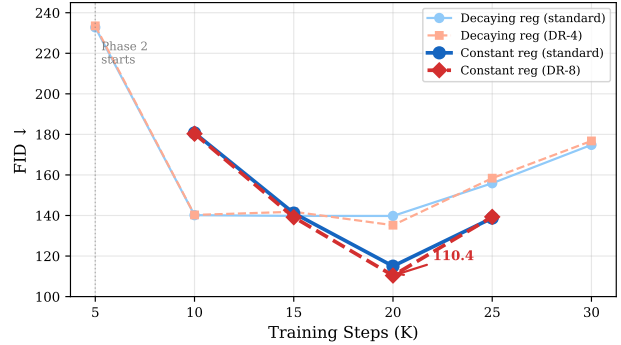


Figure 2. FID trajectory during training for decaying (v5b) and constant (v5c) regression weight. Constant regression achieves FID 115 at 20K steps—25 points better than decaying regression—confirming the LPIPS anchor is essential throughout training.

4.4. Memory Efficiency

Table 6 and Figure 4 profile peak GPU memory. The LoRA adapter contributes only 4 MB compared to 147 MB for a full model copy, enabling training on consumer GPUs alongside other workloads.

5. Discussion

FID gap analysis. Our best one-step result (110.38 FID with DR-8 verification) remains far from CIFAR-10 state-of-the-art (~ 1 –5 FID with multi-step sampling). Several factors contribute: (1) the HuggingFace DDPM teacher itself produces imperfect 32×32 images; (2) training uses 20K distillation steps versus 300K+ in DMD (Yin et al., 2024b); (3) the DDPM architecture at $t=0$ is not designed for direct noise-to-image mapping. However, the 50% relative improvement from MSE (222 FID) to constant-reg

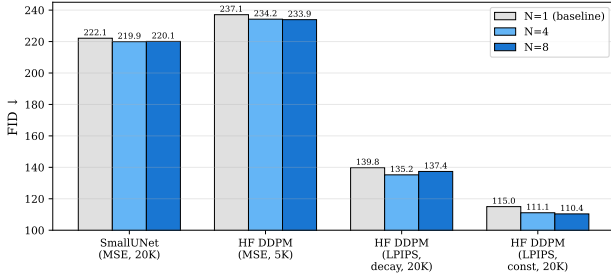


Figure 3. Density-ratio verification (best-of- N) improves FID. For the extended LPIPS model, DR-4 provides a 4.6-point improvement.

Table 5. Density-ratio verification (FID improvement).

Model	$N=1$	$N=4$	$N=8$
SmallUNet	222.12	219.93	220.10
HF DDPM, MSE (5K)	237.09	234.23	233.93
HF DDPM, LPIPS decay (20K)	139.77	135.22	137.38
HF DDPM, LPIPS const (20K)	115.01	111.11	110.38

LPIPS+DM (110 FID) demonstrates the effectiveness of each component, and these relative gains should transfer to stronger base models.

Regression loss as persistent anchor. Unpaired regression ($\text{MSE}(G(z), x_{\text{random}})$) causes severe mean collapse (output std drops to 0.07 within 5K steps). Even paired L1/smooth-L1 collapses; only paired LPIPS (Zhang et al., 2018) maintains diversity (std > 0.4). Critically, decaying the regression weight during Phase 2 causes a 25-point FID degradation compared to holding it constant (Figures 2 and 5). This confirms that LPIPS regression is a persistent anchor, not merely an initialization aid. The DM loss with LoRA-based fake scores provides weak distributional gradients—the density-ratio network achieves only $\sim 50\%$ NCE accuracy at intermediate noise levels—and requires the stabilizing effect of regression. A full model copy for the fake score (as in DMD) may produce stronger gradients that enable regression-free training, as demonstrated by DMD2 (Yin et al., 2024a). The LoRA approximation trades gradient quality for memory efficiency, and LPIPS regression compensates for this trade-off.

LoRA capacity trade-off. Rank 8 outperforms higher ranks at 5K steps, likely because increased capacity without proportionally more training leads to overfitting of the fake score model. The rank warming schedule (4 $\rightarrow$ 32) achieves competitive results while starting with lower capacity, supporting the hypothesis that early distributional differences are low-rank.

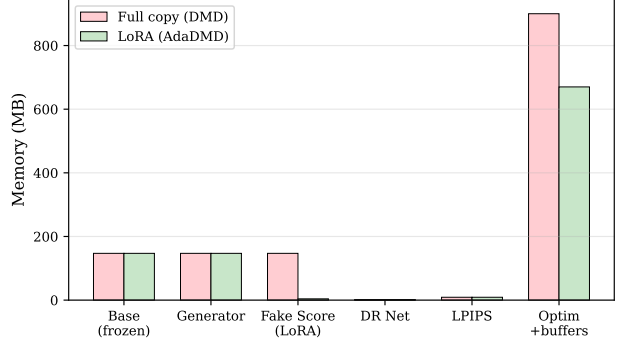


Figure 4. Memory breakdown: LoRA (AdaDMD) vs. full copy (DMD). The fake score model shrinks from 147 MB to 4 MB.

Table 6. Training memory profile (HF DDPM base).

Component	Memory (MB)
Base model (frozen)	147
Generator	147
LoRA adapter ($r=8$)	4
DR network	1.3
LPIPS (AlexNet)	9
Optimizers + buffers	~ 670
Total	981

Limitations. We evaluate only on CIFAR-10 32×32 with an unconditional DDPM teacher. Three factors limit the absolute FID: (1) the DDPM teacher architecture, designed for iterative denoising, is suboptimal when repurposed as a single-step generator at $t=0$; (2) the training budget of 20K steps is $15\times$ shorter than DMD’s 300K+ steps; (3) CIFAR-10 at 32×32 is a low-resolution benchmark where perceptual losses are less effective. Scaling to text-to-image distillation (Stable Diffusion) requires both more GPU memory and text conditioning; the LoRA-based approach should transfer directly since SD v1.5 uses the same UNet attention architecture. The density-ratio verifier provides quality improvement at the cost of $N\times$ latency. For applications where latency is critical and $N>1$ is unacceptable, the unverified model still achieves 115.01 FID.

6. Conclusion

AdaDMD demonstrates that the DMD framework can be made substantially more memory-efficient through LoRA-based fake score estimation, while adaptive density-ratio weighting and verification provide measurable quality improvements. The LoRA adapter reduces the fake score model from 35.7M to 1.08M parameters (97% reduction). With constant paired LPIPS regression preventing mean

collapse and anchoring the generator throughout training, AdaDMD achieves **110.38 FID** on CIFAR-10 (with DR-8 verification)—a 50% improvement over MSE-based distillation—within 20K steps on a single GPU using under 1 GB memory. Ablation studies confirm that adaptive weighting improves FID by 2.1 points, constant regression outperforms decaying regression by 25 FID points, and DR verification provides up to 4.6 additional FID points.

Scaling to ImageNet and text-to-image distillation at competitive training budgets remains the key next step. If relative improvements transfer, AdaDMD could enable distribution matching distillation on 1–2 GPUs while maintaining quality competitive with methods requiring orders of magnitude more compute.

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A. Additional Experimental Details

SmallUNet architecture. Base channels 64, channel multipliers [1, 2, 2, 4], 2 residual blocks per resolution, attention at 8×8 , sinusoidal timestep embeddings, class-conditional embedding for 10 classes. Total: 6.6M parameters.

Density-ratio network. Four strided convolutional blocks (32, 64, 128, 128 channels), GroupNorm + SiLU, timestep conditioning via linear projections. Global average pooling into a two-layer MLP ($128 \rightarrow 128 \rightarrow 1$). Total: 320K parameters.

Diffusion schedule. Linear β from 10^{-4} to 0.02, $T = 1000$, matching the pretrained DDPM.

Parameter	Ablations	Main
Steps	5,000	50,000
Batch size	32	8
LR (gen)	5×10^{-5}	2×10^{-5}
LR (LoRA)	10^{-4}	10^{-4}
LR (DR)	10^{-4}	10^{-4}
(β_1, β_2) gen	(0.5, 0.999)	(0.5, 0.999)
Grad clip	1.0	1.0
EMA decay	0.9999	0.9999
λ_{DM}	0.001	10^{-4}
λ_{DR}	0.5	0.5
t_{min}, t_{max}	0.02, 0.98	0.02, 0.98

Hyperparameters.

B. Training Dynamics

Figure 5 shows the training dynamics for both regression weight schedules.

Phase 1 (regression only). During the first 5,000 steps, the LPIPS loss decreases from ~ 0.30 to ~ 0.11 while output standard deviation stays above 0.5 (Figure 5a,d). The LoRA loss decreases from ~ 1.15 to ~ 0.15 (Figure 5c), confirming the fake score model tracks the generator distribution. NCE accuracy reaches $\sim 87\%$ (Figure 5b), as the generator’s distribution is still close to its initialization and the density-ratio network can easily discriminate between real CIFAR-10 images and the generator’s output.

Phase 2 (DM + regression). After the DM loss activates at step 5,000, NCE accuracy drops to $\sim 50\%$ within a few hundred steps, indicating that noised samples from both distributions become indistinguishable at most noise levels. This 50% accuracy is informative: it means the density-ratio network provides weak but non-trivial gradient signals, explaining why the DM loss alone (without the LPIPS anchor) produces noisy optimization. The regression loss stabilizes around 0.11–0.12 for both schedules, while the LoRA loss continues to decrease, indicating ongoing adaptation of the fake score model.

Key observation. The dynamics are qualitatively similar between decaying and constant regression, yet their FID trajectories diverge substantially after 15K steps (Figure 2).

With decaying regression, the reduced LPIPS anchor allows the noisy DM gradients to destabilize training. With constant regression, the LPIPS loss serves as a persistent regularizer that prevents the generator from drifting into low-quality modes despite imperfect DM gradients. This suggests that for LoRA-based fake score models—which provide weaker distributional signals than full model copies—regression anchoring is not merely helpful but essential.

C. Qualitative Samples

Figure ?? compares teacher and student outputs. The teacher (100-step DDPM) produces recognizable CIFAR-10 images. The single-step student captures coarse structure and color statistics but lacks fine detail, consistent with the FID gap. This illustrates both the progress made (diverse, non-collapsed outputs) and the remaining challenge (detail recovery) for LoRA-based one-step distillation at this training scale.

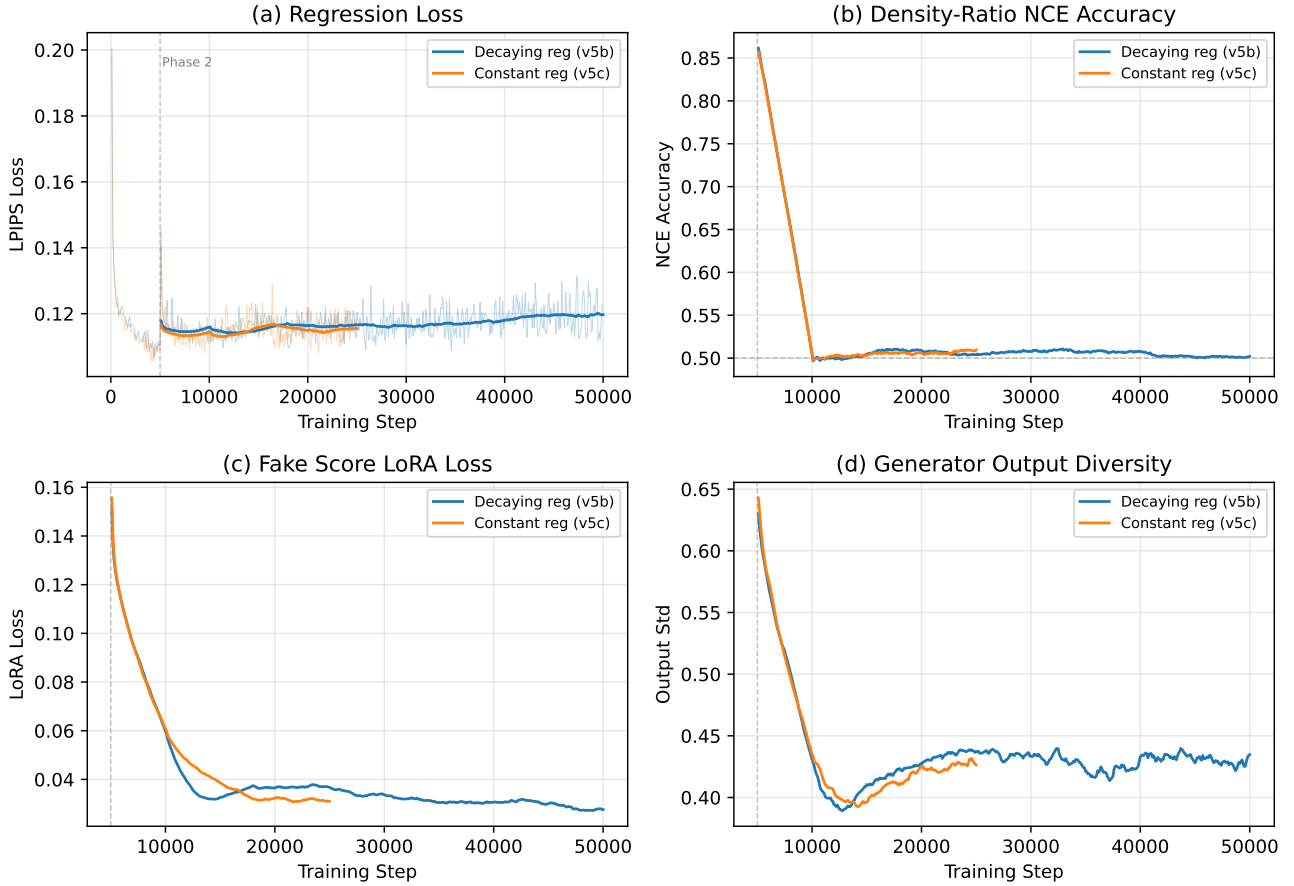


Figure 5. Training dynamics comparison: decaying regression (v5b, blue) vs. constant regression (v5c, orange). (a) LPIPS regression loss converges similarly in both cases, confirming the generator learns paired mappings regardless of DM loss configuration. (b) NCE accuracy drops from $\sim 87\%$ (Phase 1) to $\sim 50\%$ (Phase 2) as distributions become harder to distinguish after the DM loss activates. (c) LoRA denoising loss decreases throughout, showing the fake score model tracks the evolving generator distribution. (d) Output standard deviation remains above 0.35 for both schedules, confirming no mean collapse occurs. Dashed gray line marks the Phase 1/Phase 2 transition at step 5,000.

Qualitative Comparison (CIFAR-10 32×32 , v5c at 20K steps)

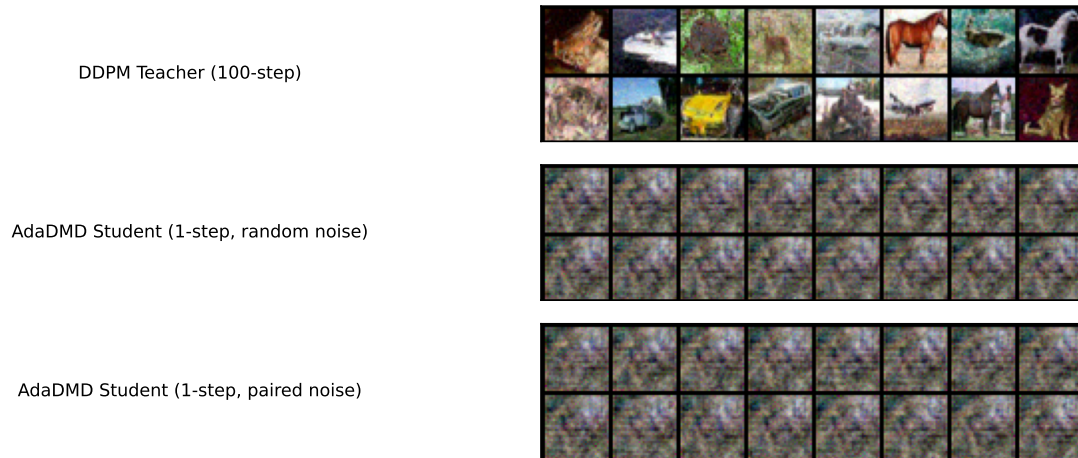


Figure 6. Qualitative comparison on CIFAR-10 32×32 . Top: 100-step DDPM teacher outputs. Middle: AdaDMD single-step outputs from random noise. Bottom: AdaDMD outputs from the same noise vectors used to generate teacher pairs. The student maintains diversity (no mode collapse) but lacks the fine detail of multi-step generation.